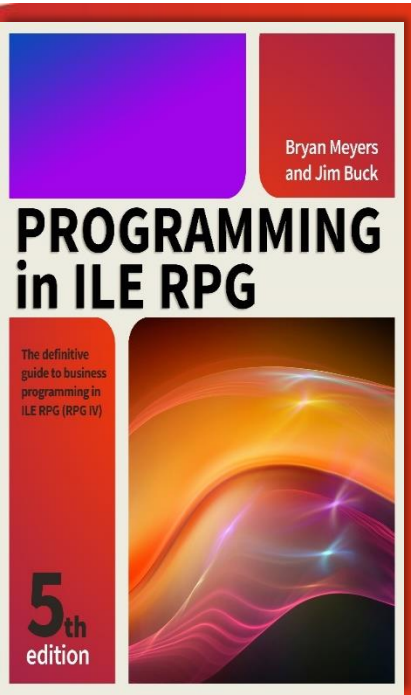
Chapter 9: Accessing and Updating Database Files



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Selected Topics

from Chapter 9 for Project 8:

Sections 9.1 – 9.2.1 Pages 259-261

Section 9.3 – 9.4.2 Pages 265-270

Sequential Access

- Program retrieves records in either arrival order—first-in, first-out (FIFO)—
  - for nonkeyed files (absence of “Keyed” on Dcl-f)
  - or in key order if file is keyed and its Dcl-f instruction indicates Keyed processing

```
Dcl-f Customers Disk Keyed;
```

  - Note that Usage(*Input) is assumed when device is Disk

Read (Read a Record)

- Retrieves next sequential record from full procedural file
  - `Read filename {data-structure};`
  - Fields in record becomes available to program as variables
  - For externally described file, can specify record format name instead of file name
- Sets %Eof (End of file) *On when no more records to read

Read (Read a Record)

- May include result data structure to receive data - Might improve performance
- Result data structure for external described file must use Likerec or Extname
  - For program described file, data structure length must match record length

```
Dcl-f Customers Disk;

Dcl-ds Custdata Likerec(Custsrec);

Read Customers Custdata;
If %Eof(Customers);
  // End-of-file processing goes here
Endif;
```

Random Access

- Retrieve specific records based on key value, without requiring sequential reading to reach that record
  - Or relative record number
  - Requires full procedural file
- `Chain` (Random retrieval from a file)

Chain (Random Record Retrieval)

- Reads record with key matching search argument

```
Chain search-argument filename {data-structure};
```

- May also include result data structure to receive data
- Turns on %Found if record is successfully retrieved
- Does not set %Eof

```
Chain Custnbr Customers;

If %Found(Customers);
  Exsr Validcust;
Else;
  Exsr Nocust;
Endif;
```

Using Composite/Concatenated Keys

- Key may consist of more than a single field
- May list key fields in parentheses
  - List may include variables, literals or expressions
  - Data class and length must correspond to key fields in file, however names do not have to match

```
Chain (Studnbr:Semester:Coursenbr) Studgrades;

Chain (Studnbr : %Char(*Year * 10 + 2) : 'SPAN312' ) Studgrades;

Chain ('346' : Region : %Char(Lastcustnbr + 1)) Customers;
```

```
// Company is “major order” key, then Region, then Custnbr
Chain (Company:Region:Custnbr) Customers;
```

Partial Composite/Concatenated Keys

- Composite/concatenated key need not include all key fields from actual file
  - If multiple records exist that match partial key, only first record is retrieved
  - Partial key must begin at major (top level) fields

Chain finds the first matching record

```
Chain (Company:Region) Customers;

If %Found(Customers);

  Dow Not %Eof(Customers);
    Exsr Process;
    Reade (Company:Region)
Customers;
  Enddo;

Else;
  Exsr Nocustomers;
Endif;
```

The use of *Reade*(Read Equal) inside this loop reads the rest of the matching records. The file must be sorted by Company and Region for this logic to work. We will study *Reade* in more depth in the advanced course.

%Kds (Key Data Structure)

- %Kds (Key data structure) function uses data structure subfields as key list

```
Dcl-ds Custkeys;
  Company Char(3);
  Region  Char(3);
  Custnbr Char(9);
End-ds;

Chain %Kds(Custkeys) Customers;
…
Chain %Kds(Custkeys:2) Customers;     // Partial composite key
```

%Kds (Key Data Structure)

- Data structure may be externally described or use Likerec

```
Dcl-ds Custkeys Ext Extname('CUSTOMERS':*Key) End-ds;

Chain %Kds(Custkeys) Customers;
```

```
Dcl-f Customers Disk;

Dcl-ds Custkeys Likerec(Custsrec:*Key);

Chain %Kds(Custkeys) Customers;
```

Legacy
“Keyed” Files – F spec entries

- File Spec entry “K”

```
FWUCRSP    IF   E                    K DISK
```

```
                File        File          End of        File
Filename        Type        Designation    File         Addition      Sequence
WUCRSP____       I           F             _            _             _
File            Limits          Record
Format      Processing      Address Type        Device
  E              _               K             DISK___

Keywords                                      Comment
```

denotes that this file has a "Key"

Legacy Code: File I/O Resulting Indicators

- File operations can set resulting indicators to return result of operation
  - Positions 71-72 = Not %Found
  - Positions 73-74 = %Error
  - Positions 75-76 = %Eof

```
*.. 1 ...+... 2 ...+... 3 ...+... 4 ...+... 5 ...+... 6 ...+... 7 ...+...
CL0N01Factor1+++++++Opcode(E)+Factor2+++++++Result++++++++Len++D+HiLoEq..

C                   Read      Customers                                8090
C     Custnbr       Chain     Customers                               7080
```